Affine-Plane Contraction Plans

Smaller Exponent Coefficients and a Joint Time–Space Certificate
for the Exact SAT Contraction Framework

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Abstract

We review twenty placed refinement packets and nine newer incoming packets in ProveIt's SubexponentialSAT project, and incorporate the newly public Anthropic formalization identified during this session. A new four-plan construction from the affine plane $\mathbb{F}_3^2$ improves the cached contraction bound. On the same generic normalized interface as the preceding coefficient comparisons, with s Boolean wires, the resulting deterministic algorithm has

$$T \le \mathrm{poly}(s+\mathcal{L})\, 2^{(30.2570356114963815\ldots+o(1))s/\sqrt{\lceil \log_2 s \rceil}},$$
$$S \le \mathrm{poly}(s+\mathcal{L})\, 2^{(25.9890677186902171\ldots+o(1))s/\sqrt{\lceil \log_2 s \rceil}}.$$

The preceding best explicit comparable cached certificate was $30.891018920366\ldots$ time and $30.753912832055\ldots$ space. The improvement is a smaller leading coefficient in an exponential upper bound; it does not change the general SAT exponent's logarithmic order. The main new ingredient is an exact joint selector controlling both the matrix-operation supports and the actual table boundaries. A rational $9 \to 12 \to 9 \to 7 \to 4$ schedule pays for recursive entry evaluations. The package supplies the proof, exact integer and rational checks, an exact selector, and finite contraction tests. It does not contain an end-to-end fast SAT solver or a new Lean formalization. The underlying deformation, packing, recovery, and SAT reductions remain explicit imported dependencies.

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1 Outcome, comparison, and scope

Let $E(s) = s/\sqrt{\lceil \log_2 s \rceil}$. The size s is the normalized Boolean wire budget of the contraction theorem, not the SAT variable count n. Initial precision and coefficient truncation are at most s, maximum root degree is at most $\left\lceil s^{9/10} \right\rceil$, and the admissible entry-access conventions are those specified in Section 3. Under these common premises our main result is

$$T \le \mathrm{poly}(s + \mathcal{L}) 2^{(C_T + o(1))E(s)}, \qquad S \le \mathrm{poly}(s + \mathcal{L}) 2^{(C_S + o(1))E(s)}, \tag{1}$$

where

$$\kappa = \frac{\pi\sqrt{2}}{\ln 2},$$

$$C_T = \kappa \left[\frac{4}{9} \sqrt{\frac{100000}{1577}} + \frac{\sqrt{137}}{90} \sqrt{\frac{25000}{303}} \right] = 30.2570356114963815\ldots, \tag{2}$$

$$C_S = \kappa \sqrt{\frac{7}{27}} \sqrt{\frac{100000}{1577}} = 25.9890677186902171\ldots. \tag{3}$$

Thus 30.26 and 26.00 are safe eventual time and space coefficients.

Comparable generic certificate	Time	Space
Best placed cached packet, *subsat_contraction_tree*	35.354588	34.274803
Newest incoming best, *sat_adaptive_tree_certificate*	30.891019	30.753913
This report, same 12/5 matrix exponent allowance	**30.257036**	**25.989068**

The numerical reductions against the incoming best are approximately 2.0523% in the time coefficient and 15.4935% in the space coefficient. They are not measured runtime or memory ratios. For example, the ratio of the two displayed leading time-bound functions is $2^{-(0.6339833088\ldots)E(s)}$; actual algorithms may run much faster than their upper bounds, and finite-size overheads are substantial.

The strongest inspected polynomial-space certificate is still 46.184845586731…, from the incoming adaptive-flags packet [6]. This report does not improve that resource regime. Keeping that algorithm and the stronger-assumption low-degree branches available preserves their guarantees. The independent Anthropic theorem has an unspecified absolute constant, so its running time cannot be numerically ranked against these wire-normalized coefficients.

What is new. The four directions in $\mathbb{F}_3^2$ give four mutually orthogonal ways to partition nine groups into three triples. Their matrix-operation supports have exact cross inner products small enough to improve the uniform adaptive bound. Tracking table-boundary supports at the same time gives a stronger memory theorem without increasing the time coefficient. The supplied selector makes this joint choice using integers. The support-averaging principle, matrix multiplication, and basic affine-plane geometry are established tools; the new contribution is their specific combination and quantitative certificate relative to the reviewed corpus. We do not claim literature-wide priority or global optimality.

What remains open. For fixed clause width, the inherited general SAT order remains

$$(n+m)^{O_k(1)}2^{O_k(n\log\log n/\sqrt{\log n})}. \tag{4}$$

The fixed-density order remains $2^{O_{k,D}(n/\sqrt{\log n})}$. No proof of a smaller logarithmic order emerged from this iteration. The conditional $n\log\log n/\log n$ proposals still lack their stated summation or representation interface. This distinction is essential to interpreting the improvement in (1).

2 Frozen review and the new Anthropic source

2.1 Repository scope

The ProveIt checkout is pinned at `73dfb5e5ab6e91bd7375e5b97ef85d7e1cc8e81d`. The relevant placement commits are `2b925d8800`, `6d3adc4506`, and `450252e9d8`; the nine newer incoming ZIPs were committed by `b9941a7396`. Both requested directories were inspected. Twenty placed packets and all nine incoming packets were screened for headline theorems and scope. Detailed proof review concentrated on the strongest generic cached and polynomial-space certificates and the interfaces used below. This is not an independent verification of every claim in all twenty-nine packets.

The original project summary still names 55.70948 as the smallest polynomial-space coefficient and 35.35459 as the smallest cached time coefficient. The newer incoming files supersede those comparison numbers. Table 1 records their generic claims, suppressing polynomial factors and the common $E(s)$ scale. A dash means that the packet does not provide a new coefficient in that column. The full source inventory and hashes are supplied separately.

Incoming packet	PS time	Cached time	Cached space
`3sat_round3_frontier`	63.75736	37.93011	33.62304
`ETH_Adaptive_Exact_Kernels`	55.33163	35.07061	33.99950
`eth_adaptive_schedules`	63.78117	37.59170	33.62304
`sat_adaptive_bounds`	**46.18485**	34.32528	34.02418
`sat_adaptive_frontier`	50.30469	31.25240	31.25240
`sat_adaptive_matrix_partition`	50.21043	34.32528	34.02418
`sat_adaptive_tree_certificate`	—	**30.89102**	30.75391
`subexponential_sat_adaptive_four`	50.39373	32.91581	31.66171
`subsat_rectangular`	53.97753	33.99389	33.99372

Table 1: Incoming claims at the frozen snapshot, using the packets' displayed generic assumptions. The four-block packet also offers a different matrix backend; that optional point does not beat the highlighted incoming time.

The strongest density-sensitive order retained in the portfolio is

$$(n+m)^{O_k(1)}2^{O_k\left(\frac{n}{\sqrt{\log n}}\min\{\log(2+m/n),\log\log n\}\right)}, \tag{5}$$

in the parameter ranges established by the two density packets [7]. This improves some growing-density regimes, not the unrestricted worst-case order. All comparisons with (5) keep its hypotheses and normalization separate from (1).

2.2 Publication update supplied during the session

The user supplied Anthropic pull request 31 at exact revision `1a6eee293b9e00cb432eef6018a21f523623a9af` [8]. It was merged on 26 October 2026 at 8:07:22 p.m. Pacific time. This changes an

outdated statement in the earlier ProveIt reports: the Claude paper's Lean development is now publicly available.

The published `eth/MainResults.lean` defines one finite-alphabet deterministic Mathlib TM2 machine and one constant C, both chosen before k, with counting time

$$2^{C\min\{km/\sqrt{\log_2 m},\, k^2 n \log_2\log_2 n/\sqrt{\log_2 n}\}}. \tag{6}$$

Its explicit premises are $k \geq 1$, $m \geq 2$, $n \geq 4$, no repeated clause lists, and occurrence of every named variable. The source also gives a separate ETH-negation statement. `Solution.lean` contains proof terms for both. `Challenge.lean` contains the intentional comparison stubs; those stubs are not the proof terms in `Solution.lean`.

The README pins Lean and Mathlib v4.33.1 and gives build, axiom-inspection, Comparator, and independent-kernel instructions. We inspected the statements, entry points, paper, and source index; we did not rebuild the Lean development. Accordingly this report attributes the formal verification claims to the published project and does not report a new local proof-assistant check. The new 49-page PDF differs in bytes from the older 49-page ProveIt copy; its SHA-256 and the comparison are recorded in the package. Its headline order is still (6).

The Anthropic construction offers a distinct research interface, based on integral decompositions of rectangle swaps and choosing a cheaper row or column contraction. Section 10 discusses its implication for the next order-improvement attempt. It is not used as an unproved shortcut in our new coefficient theorem.

3 The normalized contraction interface

We import the exact arithmetic and reduction framework of the OpenAI manuscript and the compatible refinements [2, 3, 4, 5]. The root has at most s wires and entries in

$$(\mathbb{Z}/2^{P_0}\mathbb{Z})[V]/(V^{\ell}), \qquad P_0, \ell \leq s,$$

maximum degree at most $\left\lceil s^{9/10} \right\rceil$, and total description length $\mathcal{L}$. Admissible original entry routines use polynomial space and $\mathrm{poly}(s+\mathcal{L})2^{O(s^{9/10})}$ time with fixed constants. The extension ring, canonical lifts, and immutable creation-precision tags are retained. In particular, a recursively created tensor is evaluated from its fixed creation-time representatives, rather than by recursively rerunning its whole ancestor contraction.

At an active call let M be the current wire count, r the fixed number of groups, u the retained interpolation range, a the cumulative packet exponent, and $D = 2^{a-i}$ at retained valuation $i < u$. The inherited mechanisms give near-unit padding, child size $(2/r + o(1))M$, working precision $(6 + o(1))MD/u$, and full-Cauchy modes with weights

$$a + 1 - i + 4q - 2\,\mathrm{popcount}(q), \qquad q \geq 0.$$

There are at most $(2 + o(1))M/(r^2 D)$ families per directed group pair and index. These are mathematical dependencies, not newly proved facts here.

For each directed pair $g \neq h$, allocate an exact nonnegative mode weight b_{gh}, with total below the working precision. Let $\mathcal{I}_{gh}$ be the corresponding finite list. An empty list makes that allocation zero; otherwise write $h_{gh} = \log_2 |\mathcal{I}_{gh}|$. Its contribution is the finite graphical sum

$$\sum_{\xi_{gh} \in \mathcal{I}_{gh}} \prod_{g=1}^{r} Z_g((\xi_e)_{e \ni g}), \tag{7}$$

where evaluating an entry of Z_g requires a real recursive contraction. Exact list cardinalities and indexing are obtained by the ordinary unbounded-knapsack recurrence. As r ranges over a fixed finite set, the number of exact-weight allocations is polynomial in s.

Lemma 3.1 (Imported uniform entropy estimate)**.** *Put $\mathcal{K}(M,u) = \kappa M/\sqrt{u}$, with $\kappa = \pi\sqrt{2}/\ln 2$. For every member of a fixed finite family of nonnegative directed weights $z = (z_{gh})$,*

$$\sum_{g\neq h} z_{gh}h_{gh} \le \left(\frac{\|z\|_2}{r} + o(1)\right)\mathcal{K}(M,u).$$

The error is uniform over the active depths and exact-weight allocations of our schedule.

Source and normalization. The full-Cauchy generating-function estimate is

$$\sum_{q\ge 0} -\ln(1-e^{-\theta(1+4q-2\,\mathrm{popcount}(q))}) = \frac{\pi^2}{24\theta} + O(\log^2(2/\theta)).$$

For pair e use parameter θ/z_e, multiply its logarithmic bound by z_e, and sum. The leading expression is $\theta p + \pi^2(2+o(1))M\sum_e z_e^2/(24\theta r^2 D)$. Using $p \le (6+o(1))MD/u$ and optimizing θ gives the displayed constant after division by $\ln 2$. The additive precision guard makes the error lower order uniformly. This reproduces the normalization of the inherited estimate; its mode factorization and precision premises are imported from the cited packets. □

For an unordered pair $e = \{g,h\}$ define $g_e = h_{gh} + h_{hg}$. Thus for any fixed nonnegative unordered support w,

$$w\cdot g \le \left(\frac{\sqrt{2}\,\|w\|_2}{r} + o(1)\right)\mathcal{K}(M,u). \tag{8}$$

The factor $\sqrt{2}$ matters: a support on an undirected pair weights both orientations in the underlying list representation.

4 Operation and storage supports

A plan partitions the groups into three nonempty ordered blocks. Within each block it recursively splits an ordered list at its midpoint and contracts the two resulting tables. It then contracts the final triangle of three block tables. There are $r-3$ internal merges and one final triangle. A subset table retains precisely the indices on its boundary.

For an internal merge, write the three disjoint vertex classes as A, B, O, where O is the complement of $A\cup B$. Let H_1, H_2, H_3 be the sums of g_e on its three crossing pair classes. With the same conservative matrix exponent allowance $\omega < 12/5$ used in the incoming best, rectangular blocking gives operation time exponent

$$H_1 + H_2 + H_3 - \frac{3}{5}\min(H_1,H_2,H_3) = \max_j \left(\frac{2}{5}H_j + \sum_{k\neq j} H_k\right). \tag{9}$$

The final triangle has the same three forms. Multiplying by five gives integer support vectors $v \in \{0,2,5\}^{\binom{r}{2}}$. For plan j, let $\mathcal{V}_j$ be its $3(r-2)$ supports and put

$$F_j(g) = \frac{1}{5}\max_{v\in\mathcal{V}_j} v\cdot g.$$

For every table subset S let b_S be the 0–1 indicator of its unordered boundary. Include singleton input caches, internal subset tables, and final block tables, and put

$$G_j(g) = \max_{S \text{ in plan } j} b_S \cdot g.$$

The final product of two block tables has the boundary of the complementary third block, so it is covered by the same list.

Lemma 4.1 (Arithmetic and workspace)**.** *After the singleton factor entries have been evaluated, a plan can be executed with local time* $\mathrm{poly}(s)2^{F_j(g)}$ *and space* $\mathrm{poly}(s)2^{G_j(g)}$ *over the inherited coefficient ring.*

Proof. For matrix dimensions a, b, c, divide into square blocks of side $m = \min(a, b, c)$. A fixed bilinear multiplication algorithm of exponent below $12/5$ gives $abc\,m^{-3/5}$ work up to polynomial factors in record length, yielding (9). Such algorithms exist over characteristic zero with a strict exponent margin [9]. The inherited coordinate/CRT bridge expands each ring entry into polynomially many integer coordinates, uses sufficiently many nonexceptional odd primes, reconstructs exact bounded integer matrix products, applies the ring structure constants, and reduces to the desired precision. It never divides by a nonunit of the original ring. All coordinate counts and reconstruction bit lengths are polynomial in s.

Workspace deserves a separate argument. Implement the fixed bilinear algorithm recursively, processing its fixed number of subproducts sequentially and accumulating into the output. The work arrays at successive sizes have a geometrically decreasing sum, so square multiplication uses $O(m^2)$ scalar storage. Process rectangular blocks and CRT coordinates sequentially. Storage is a polynomial factor times the input and output matrix entries. Those matrices are exactly the subset boundary tables counted by G_j, or submatrices of them. At most a constant number of tables per plan coexist, since r is fixed. This proves the stronger boundary-based space bound; merely bounding storage by time would lose the improvement below. □

The proof relies on the stated theoretical matrix backend. Our finite contraction implementation uses ordinary exact scalar arithmetic to test the regrouping, not an implementation of that asymptotic backend.

5 Four plans from the affine plane

Label the nine groups by $\mathbb{F}_3^2$, using labels $3x + y$ for $x, y \in \{0, 1, 2\}$. The level sets of four projectively distinct nonzero linear forms give four partitions into parallel lines. Use the following ordered triples; their order also specifies the internal midpoint splits:

Plan	Three final blocks
0	$(0,1,2) \mid (3,4,5) \mid (6,7,8)$
1	$(0,3,6) \mid (1,4,7) \mid (2,5,8)$
2	$(0,5,7) \mid (1,3,8) \mid (2,4,6)$
3	$(0,4,8) \mid (2,3,7) \mid (1,5,6)$

Any line from one partition meets any line from another in one point. Every unordered pair is internal to one of the four partitions and crosses the other three. These two exact incidence properties drive the bound.

Each plan has twenty-one operation supports and fifteen stored-table boundary supports: nine singletons, three two-point subsets, and three lines. We use both support families in the selector.

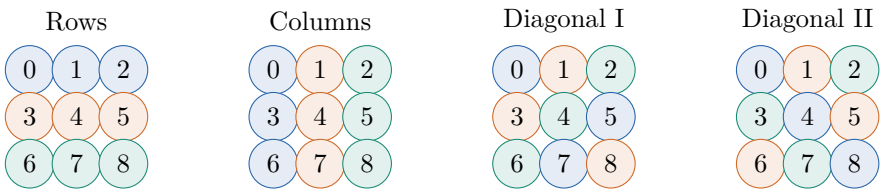


Figure 1: The same nine groups under four partitions. Each color class is one three-group block; the diagram displays finite-field classes, not Euclidean line segments.

Lemma 5.1 (Exact affine support bounds)**.** *For a time support $t = v/5$ and a boundary support b of any plan,*

$$\|t\|_2^2 \le \frac{486}{25}, \qquad \|b\|_2^2 \le 18.$$

For supports belonging to distinct affine directions,

$$t \cdot t' \le \frac{288}{25}, \qquad b \cdot b' \le 8, \qquad t \cdot b' \le \frac{48}{5}. \tag{10}$$

Proof. For a time operation the three class sizes are $(1,1,7)$, $(1,2,6)$, or $(3,3,3)$. Its maximum integer squared norm is respectively 354, 458, or 486. This follows by assigning the discounted coefficient two to the smallest crossing class and coefficient five to the other classes.

For two directions the 3×3 array of intersections has one point per cell. Enumerating the three possible discounted pair classes in each operation gives the following upper bounds on integer inner products:

	$(1,1,7)$	$(1,2,6)$	$(3,3,3)$
$(1,1,7)$	204	208	215
$(1,2,6)$	208	240	260
$(3,3,3)$	215	260	288

These finite counts can be reconstructed directly by assigning each pair of labels its operation classes. The independent verifier supplied with this paper enumerates all $6 \cdot 21^2 = 2646$ cross-direction time pairs, all self norms, and both mixed families, using only integer membership tests. Thus the table is a complete finite certificate, not a sample.

There is also a useful direct explanation of the largest entry. A final time support is the indicator of the twenty-seven crossing pairs minus $3/5$ times the indicator of one discounted nine-pair class. Across two directions, the full crossing sets meet in eighteen pairs, a discounted class meets the other's full crossing set in six, and the two discounted classes meet in two. Their inner product is therefore $18 - (3/5)12 + (9/25)2 = 288/25$, independently of the chosen discounts.

For boundaries of subsets S, T of sizes $a, b \le 3$, contained in lines of different directions, $c = |S \cap T|$ is zero or one. Their boundary intersection has size

$$c(9 - a - b + c) + (a - c)(b - c).$$

If $c = 1$, this is $(a-2)(b-2) + 7 \le 8$. If $c = 0$, it is ab; $a = b = 3$ is impossible because the two full lines meet, so it is at most six. Also $|\partial S| = a(9-a) \le 18$. Finally, enumerating the same 3×3 intersections for a time operation and a boundary gives $48/5$. For a final time operation and a line boundary this is $12 - (3/5)4 = 48/5$; smaller subsets and internal operations are no larger, as the complete finite check confirms. □

Corollary 5.2 (Affine local time)**.** *Selecting the least-time affine plan gives local aggregation exponent at most* $(1/\sqrt{3}+o(1))\mathcal{K}(M,u)$.

Proof. Choose a maximizing support in each of the four plans. The squared norm of their sum is at most $4(486/25)+12(288/25)=216$. The minimum of four maxima is at most their average. Apply (8) to the average support: its squared norm is at most $216/16$, and $\sqrt{2(216/16)}/9=1/\sqrt{3}$. Equality in the support-norm certificate is attained by any choice of four final-triangle supports. This establishes the certificate value, not a global algorithmic optimality claim. □

6 A joint selector improves memory as well

Minimizing time and minimizing storage independently would select potentially different plans. It would not establish a simultaneous time–space pair. The following selector resolves this issue explicitly.

Theorem 6.1 (Joint affine time–space certificate)**.** *Set* $\rho=3/\sqrt{7}$. *Select a plan minimizing an upward-rounded proxy for*

$$J_j(g)=\max\{F_j(g),\rho G_j(g)\}.$$

Then, uniformly over the feasible mode allocations,

$$F_j(g)\le\left(\frac{1}{\sqrt{3}}+o(1)\right)\mathcal{K}(M,u),\qquad G_j(g)\le\left(\sqrt{\frac{7}{27}}+o(1)\right)\mathcal{K}(M,u).$$

The selection uses exact integers and polynomial overhead.

Proof. For each plan the support family of J_j is the union of its time supports $v/5$ and its scaled boundary supports ρb_S. Choose a maximizing support from each plan. If k of the four are time supports, the squared norm of their sum is at most

$$B_k=k\frac{486}{25}+k(k-1)\frac{288}{25}+\rho^2\big[18(4-k)+8(4-k)(3-k)\big]+2k(4-k)\rho\frac{48}{5}.$$

At $\rho=3/\sqrt{7}$ this expression has the exact identity

$$B_k=216-8k(4-k)\left(\frac{6}{5}-\frac{3}{\sqrt{7}}\right)^2\le 216. \tag{11}$$

Averaging as in Corollary 5.2 therefore bounds $\min_j J_j$ by $(1/\sqrt{3}+o(1))\mathcal{K}$. The support family is fixed and finite, so the inherited entropy error remains uniform despite the adaptive choice.

For exact selection, let $d_{\{g,h\}}=|\mathcal{I}_{gh}|\,|\mathcal{I}_{hg}|$ and compute

$$A_j=\max_{v\in\mathcal{V}_j}\prod_e d_e^{v_e},\qquad B_j^{\mathrm{tab}}=\max_S\prod_e d_e^{b_S(e)}.$$

Let $A_j^+=\lceil\log_2 A_j\rceil$ and $B_j^+=\left\lceil\log_2 B_j^{\mathrm{tab}}\right\rceil$. They are computed by `(P-1).bit_length()` for positive integer P, including $P=1$. Comparing the integers

$$\max\{7(A_j^+)^2,225(B_j^+)^2\} \tag{12}$$

is exactly equivalent to comparing $\widehat{J}_j = \max\{A_j^+/5, 3B_j^+/\sqrt{7}\}$: the displayed score is $175\widehat{J}_j^2$. Moreover $J_j \le \widehat{J}_j < J_j + \rho$ unless all relevant costs are exact, in which case the non-strict upper bound still holds. The selected plan therefore obeys $J_j \le \min_l J_l + \rho$. This additive constant in the local exponent changes neither leading coefficient. Use the first plan in a tie. All counts, products, bit lengths, and scores have polynomial bit length. No floating-point or irrational comparison enters the algorithm. □

The new storage coefficient is $\sqrt{7/27} = 0.509175\ldots$ locally, compared with the time coefficient $1/\sqrt{3} = 0.577350\ldots$. Considering boundaries alone gives the same squared coefficient $7/27$ from $4 \cdot 18 + 12 \cdot 8 = 168$. Thus simultaneous time control incurs no leading space loss relative to this marginal support certificate.

Proposition 6.2 (Scope of local sharpness)**.** *The coefficients $1/\sqrt{3}$ for time and $\sqrt{7/27}$ for space are individually sharp for these four specified plans under only the Euclidean pair-entropy relaxation and the stated operation/table cost model.*

Proof. For time, choose a final-triangle integer support from each plan and set g equal to their sum. Then $\|g\|_2^2 = 5400$ and each chosen support has inner product $486 + 3 \cdot 288 = 1350$ with g. Every plan thus has time cost at least 270, attaining the upper certificate after normalization. For space choose in each direction the line through zero and let g be the sum of their four boundary indicators. Each boundary has squared norm 18 and every cross inner product is eight. Hence $\|g\|_2^2 = 168$ and each chosen boundary has g-weight $18 + 3 \cdot 8 = 42$. This attains the space coefficient under (8). These nonnegative profiles are witnesses for the finite cost model. We do not assert that they arise from the exact mode lists of a SAT instance, or infer a SAT runtime or general memory lower bound. □

7 A twelve-group certificate for the second level

The old $9 \to 13 \to 7$ schedule is limited by the cost of caching the root factors and then doing the depth-one contraction. Improving the root operation alone would not improve its displayed time coefficient. We therefore also need a smaller second-stage certificate and a fully paid schedule. The following four ordered plans on twelve groups suffice:

Plan	Three final blocks
0	$(11,7,2,0) \mid (5,9,4,3) \mid (6,1,10,8)$
1	$(3,9,6,0) \mid (10,1,4,7) \mid (2,8,5,11)$
2	$(0,10,4,5) \mid (11,3,8,1) \mid (6,9,2,7)$
3	$(1,5,7,6) \mid (3,0,10,2) \mid (9,11,4,8)$

Each plan has thirty integer operation supports. The exact certificate is

$$\max_{v_j \in \mathcal{V}_j} \|v_0 + v_1 + v_2 + v_3\|_2^2 = 9864, \qquad \beta_{12}^2 = \frac{2 \cdot 9864}{25 \cdot 16 \cdot 12^2} = \frac{137}{400}. \tag{13}$$

The supplied checker enumerates all $30^4 = 810000$ support tuples with integer arithmetic. Precomputed pairwise inner products accelerate this exact enumeration; the certificate is the fixed menu and exhaustive maximum, not a numerical search result. The least-time plan consequently has local time and space coefficient $\beta_{12} = \sqrt{137}/20$ by the support-averaging theorem of the prior packet [5]. Space here may be bounded by operation cost: expression (9) dominates every input/output table size.

For the small alphabets $r = 7$ and $r = 4$, retain fixed balanced plans with final block sizes $(3, 2, 2)$ and $(2, 1, 1)$. Their largest directed support norms give

$$\beta_7^2 = \frac{632}{1225}, \qquad \beta_4^2 = \frac{13}{25}. \tag{14}$$

At both occurrences of $r = 9$ the affine method is available; the sharper space selector is required at the root, and may also be used later.

8 The complete paid recursion

8.1 Recurrence and the cost of tensor entries

Every group factor in (7) depends on $2(r-1)$ directed lists. The entropy estimate bounds the number of its entries by $2^{(\alpha_r+o(1))\mathcal{K}(M,u)}$, where

$$\alpha_r = \frac{\sqrt{2(r-1)}}{r}.$$

Each entry is computed using the actual child contraction. Combining this with the local support theorem gives

$$T_d \le \mathrm{poly}(s)\left[2^{(\alpha_{r_d}+o(1))\mathcal{K}_d} T_{d+1} + 2^{(\beta_{r_d}+o(1))\mathcal{K}_d}\right], \qquad \mathcal{K}_d = \kappa M_d/\sqrt{u_d}. \tag{15}$$

At $r = 9$, $\beta_9 = 1/\sqrt{3}$ even with the joint selector. The α term cannot be discarded or replaced by a unit-cost entry oracle.

8.2 Integer schedule

Let $\lambda = \lceil \log_2 s \rceil$, $\Lambda = \lambda - \left\lceil \sqrt{\lambda} \right\rceil$, and $B = \lceil 6 \log_2 \lambda \rceil$. Set

Depth d	r_d	η_d
0	9	1577/100000
1	12	303/25000
2	9	73/20000
3	7	1/250
$4+j$, $j \ge 0$	4	$\frac{26751}{5000000}(63/100)^j$

Use

$$u_d = \lfloor \eta_d \Lambda \rfloor, \qquad a_d = \sum_{j=0}^{d} u_j + (d+1)B, \qquad H = \lceil s/\lambda \rceil. \tag{16}$$

Calls with $M \le H$ are evaluated directly. The tail budget is 723/50000 and the full leading budget is exactly

$$\sum_{d \ge 0} \eta_d = \frac{1}{20}. \tag{17}$$

Use near-unit padding with slack λ^{-3}. Handle a fixed finite exceptional range by direct contraction. All required inequalities are eventually valid, as the following argument establishes.

8.3 Packing, depth, and precision gates

Write $A_d = \sum_{j \le d} \eta_j$. At the root, the dominant packing exponent is $9/10 + 2A_0 \log_2 r_0$. At later calls it is $9/10 + A_d \log_2 r_d$, because the retained side may still have root degree, while the fresh side has degree at most $2^{a_{d-1}+1}$. Exact integer powers certify

$$\log_2 9 < 317/100, \quad \log_2 12 < 717/200, \quad \log_2 7 < 281/100.$$

Using these rational upper bounds gives the following positive gaps below one at the four prefix depths:

$$\frac{91}{5000000}, \quad \frac{287}{20000000}, \quad \frac{91}{5000000}, \quad \frac{663}{5000000}.$$

In the four-group tail the limiting retained-side exponent reaches one at the leading boundary. The retreat from λ to Λ gives the explicit correction $-\left\lceil \sqrt{\lambda} \right\rceil /10 + O((\log \lambda)^2)$ to its binary logarithm, which absorbs every guard and polylogarithmic factor. The fresh-side and packet-rounding exponents are at most $1/4 + o(1)$. New packet degrees are at most $s^{1/20+o(1)}$, below the root degree allowance.

Here is the noncircular depth argument. Consider hypothetical active depths up to $\log_{20/11} \lambda + O(1)$, with $M > H$. Throughout that range the smallest taper is

$$\lambda^{1+\ln(63/100)/\ln(20/11)+o(1)},$$

which tends to infinity because $63/100 > 11/20$. The accumulated guard is $O((\log \lambda)^2)$, and the preceding margins hold uniformly. Consequently the tail child ratio is at most $11/20$ eventually, so an active path cannot reach the hypothetical limit. This closes the depth bootstrap. The padding errors accumulated over $O(\log \lambda)$ levels are $1 + o(1)$, and floors are uniformly negligible.

The root and prefix wire masses are therefore at most $(1 + o(1))s$ times

$$c_0 = 1, \quad c_1 = \frac{2}{9}, \quad c_2 = \frac{1}{27}, \quad c_3 = \frac{2}{243}, \quad c_4 = \frac{4}{1701}, \tag{18}$$

followed by a factor $1/2$ per tail depth. For retained $i < u_d$, the quantity $D = 2^{a_d - i}$ is at least $2^{a_{d-1}+B+1}$. The inherited precision divided by MD/u_d is bounded by $O(\lambda^4 2^{-B}) = o(1)$, so the sharp local precision and entropy bounds remain valid. All arithmetic bit lengths are polynomial in s. Direct leaf work $2^{O(s/\lambda)}$, original entry work $2^{O(s^{9/10})}$, and fresh packet-entry work $\exp(s^{1/20+o(1)})$ are lower order on $E(s)$.

8.4 Exact maximization of the paid branches

Put $K_d^* = \kappa c_d / \sqrt{\eta_d}$ for $0 \le d \le 4$ and

$$\gamma = \frac{1/2}{\sqrt{63/100}} = \frac{5}{\sqrt{63}} < 1.$$

Beyond depth four the K_d^* form a geometric sequence with ratio γ. For a local aggregation at depth d, unrolling (15) gives $\sum_{j<d} \alpha_{r_j} K_j^* + \beta_{r_d} K_d^*$. The tail expressions are affine in γ^j; hence their supremum is covered by the first tail aggregation and the infinite terminal cache chain. Exact outward rational interval arithmetic gives:

Paid branch	Coefficient on $E(s)$
Root aggregation	29.46883285288760
Depth-one aggregation	**30.25703561149638**
Depth-two aggregation	30.01092197330065
Depth-three aggregation	30.08781178892023
First tail aggregation	30.05006255795701
Infinite terminal cache chain	30.24246506295926

The depth-one branch gives (2). All leaves and child calls have been included. Products of polynomial overheads over $O(\log \lambda)$ depths add $O(\lambda \log \lambda) = o(E(s))$ to the exponent. The input description has one fixed polynomial dependence on $s + \mathcal{L}$, as in the inherited entry interface.

8.5 Space is a maximum, not a sum of exponents

At the root, Theorem 6.1 gives coefficient $\sqrt{7/27}\, K_0^* = 25.9890677186902171\ldots$. Safe upper bounds at the next four stages are respectively 7.5719175044, 2.2686556743, 0.5991335376, and 0.1485975848. The tail then decreases geometrically. Computing a child while storing partially filled ancestor tables adds their storage sizes; it does not multiply them. Only $O(\log \lambda)$ levels are simultaneously active. Lemma 4.1 and polynomial record lengths consequently give the maximum of these local space exponents, attained at the root. This proves (3) and the main bound (1).

9 Reproducibility and the proof boundary

The supplied software is standard-library Python. It uses explicit checks that remain active under `python -O`. The principal verification layers are:

1. Independent reconstruction of the four affine plans, all operation and boundary supports, self norms, cross inner products, and joint k-type inequalities. Deliberately replacing an affine direction by a duplicate direction is rejected.

2. Exhaustive verification of the twelve-group finite certificate, including every one of its 810000 support tuples. Its exact integers are supplied with the fixed plan data.

3. Exact evaluation of the rational schedule, logarithm gates, geometric tail, and all paid branches. Square roots are bounded by integer arithmetic; π uses a rational Machin-series enclosure and $\ln 2$ a positive convergent series with a rational tail bound. Incorrect time, capacity, and tail claims are rejected by negative controls.

4. Exact plan selection and small graphical contractions, compared with literal enumeration over finite rings containing zero divisors. These checks validate reassociation and indexing, not the theoretical fast-matrix running time.

Normal and optimized replays produce matching mathematical outputs. The artifact records executed results and SHA-256 identities. Hashes establish byte consistency, not proof validity. The natural-language proofs and exact finite certificates do not replace a formalization of the imported contraction framework. No full fast SAT implementation, SAT runtime benchmark, or new Lean/Rocq theorem is claimed.

To replay from the extracted archive:

```
python3 code/run_checks.py
python3 -O code/run_checks.py
```

To rebuild the report, with a normal TeX Live or MiKTeX installation:

```
python3 code/build_report.py
```

Neither command needs network access. The build disables shell escape. The package is intended as reviewable research material for later incorporation into ProveIt; no remote branch or repository file was modified during this session.

10 What would change the logarithmic order?

10.1 What this construction can and cannot change

The affine construction improves the constant multiplying the entropy scale $\mathcal{K}(M,u) = \Theta(M/\sqrt{u})$. It does not replace that scale. The packing constraints still force the root taper to be $O(\log s)$. Moreover, an equal positive pair-entropy profile forces every complete subset-table plan through a boundary of quadratic size in the group count. This is a barrier for this explicit representation and analysis, not a lower bound for SAT. It explains why merely increasing the number of adaptive plans cannot be presented as a proof of a $1/\log n$ denominator.

The adaptive-flags packet proves its binary balanced-tree local bound sharp under the Euclidean entropy information it uses. We also examined ordinary schedule retuning in that regime; it does not provide a substantive new polynomial-space result. The successful change here is to the cached plan geometry and simultaneous memory analysis.

10.2 The newly public swap interface

The Anthropic paper makes a separate target concrete. Its integral crossing decompositions control a simultaneous row/column boundary tradeoff, and its quantitative recursion uses a parameter $\rho = 1/\sqrt{2}$. In its stated interface the logarithmic-denominator power is $\log_2(1/\rho)$; this gives $1/2$ at the present value. A compatible decomposition with strictly smaller ρ, including degree, integrality, construction-time, and entry-access bounds, would therefore constitute an order improvement within that framework.

The paper's Appendix H already discusses improving the graded-dimension growth that controls this parameter. That observation is prior work in the newly supplied source, not a new result of this report. Our affine plan proof is about finite support geometry after mode lists have been formed; it does not establish a better swap decomposition or a smaller ρ. Mixing the numerical constants of the two interfaces without proving a translation would be unjustified.

A useful next experiment is an exact search for small integral crossing decompositions, tracking both orientation ranks and coefficient/degree growth simultaneously, followed by a proof that the constructions compose. Such a search needs negative examples and exact rank witnesses. It should be evaluated against the public Definition 7/Theorem A interface, rather than against a single informal rank estimate. We leave this as an explicit research objective; the completed deliverable is the smaller joint time–space certificate proved above.

A Claim and dependency ledger

Item	Status and scope
Four affine plans	New construction in this reviewed corpus; exact finite support proof and independent integer checks.
Joint integer selector	New simultaneous time–space certificate; actual table boundaries and matrix workspace are included.
Twelve-group menu	New fixed finite certificate; complete 30^4 check.
Rational schedule	New compatible composition; all prefix, tail, and child evaluation costs included.
C_T, C_S	Proved relative to the imported normalized interface; exact rational interval enclosures supplied.
General SAT order	Unchanged; no new n-only or fixed-density logarithmic order is established.
Matrix backend	Imported theoretical result and exact coordinate/CRT bridge; finite code uses naive multiplication.
Deformation and recovery	Imported OpenAI manuscript identities and previous ProveIt refinements.
Mode entropy and packing	Imported full-Cauchy weighted count and asymmetric near-unit padding; premises checked for the new schedule.
Anthropic source	Public statement/source review at the exact user-supplied commit; no local Lean rebuild.
Runtime experiments	Finite correctness/certificate computations only; no measured practical SAT speedup.
Novelty	Relative to the screened twenty-nine packets; no exhaustive literature priority claim and no global optimum claim.

B How the source ledger is organized

`provenance/source_inventory.json` records exact repository pins, selected source paths, Git blob identities where available, archive sizes, and SHA-256 hashes. `provenance/review_inventory.csv` gives one row for each of the twenty-nine packets. The companion Anthropic update identifies the old and new manuscript bytes and the exact statement files read. The archive intentionally contains our deliverable and concise provenance rather than duplicate copies of every upstream paper and repository.

References

[1] V. Reshetnikov, *ProveIt: SubexponentialSAT and incoming research*, snapshot `73dfb5e5ab6e91bd7375e5b97ef85d7e1cc8e81d`. https://github.com/VladimirReshetnikov/ProveIt/tree/73dfb5e5ab6e91bd7375e5b97ef85d7e1cc8e81d/Computability/SubexponentialSAT.

[2] OpenAI, *The Exponential Time Hypothesis is False*, 25 October 2026. Vendored sources pinned to upstream `7ef5c0cc5ca4b0e072b7a050fca09ab24bc6fa3a`, especially `build/quantitative-contraction.tex` and `build/quantitative-reduction.tex`. https://github.com/openai/math/tree/7ef5c0cc5ca4b0e072b7a050fca09ab24bc6fa3a/preprints/The-Exponential-Time-Hypothesis-is-False-October-25-2026.

[3] *ETH asymmetric packet balancing*, 26 October 2026. ProveIt, `Computability/SubexponentialSAT/Research/eth_asymmetric_packet_balancing/`.

[4] *Joint Cauchy contraction* and *Adaptive aggregation for subexponential SAT*, 26 October 2026. ProveIt research packets `eth_joint_cauchy/` and `sat_adaptive_contraction/`.

[5] *Adaptive contraction trees with exact support certificates*, 26 October 2026. Incoming archive `sat_adaptive_tree_certificate_2026-10-26.zip`, arrival commit `b9941a7396e465f41df38326284d0500a117d775`. The report's support-averaging theorem and $9 \to 13 \to 7$ paid recurrence are the principal cached comparison.

[6] *Adaptive contraction beyond the October 26 SAT frontier*, 26 October 2026. Incoming `sat_adaptive_bounds_2026-10-26.zip`; the complete binary linear-flag theorem and its polynomial-space schedule.

[7] *Density-aware subexponential SAT improvement* and *ETH density refinement*, 26 October 2026. ProveIt research packets `sat_density_improvement/` and `eth_density_refinement/`.

[8] Claude (Anthropic), *A $2^{o(n)}$ Time Algorithm for 3-SAT*, paper and Lean project, exact revision `1a6eee293b9e00cb432eef6018a21f523623a9af`. https://github.com/anthropics/formal-math/pull/31. Statement: https://github.com/anthropics/formal-math/blob/1a6eee293b9e00cb432eef6018a21f523623a9af/eth/MainResults.lean. Proof entry point: https://github.com/anthropics/formal-math/blob/1a6eee293b9e00cb432eef6018a21f523623a9af/eth/Solution.lean.

[9] V. Vassilevska Williams, Y. Xu, Z. Xu, and R. Zhou, *New Bounds for Matrix Multiplication: from Alpha to Omega*, SODA 2024. https://arxiv.org/abs/2307.07970. The stated bound $\omega \leq 2.371552$ is more than sufficient for the strict $12/5$ allowance retained here; no claim about the latest optimal matrix exponent is needed.